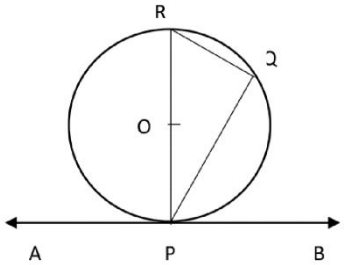
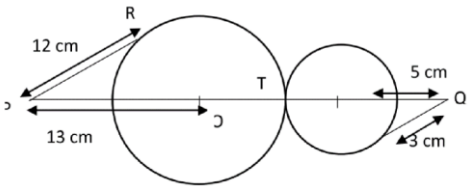
CIRCLES

1. In the figure, AB is a tangent to a circle with center O. Prove that ∠BPQ = ∠PRQ.
If ∠BPO = 60°, find ∠RPO.

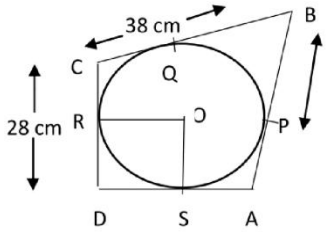


2. Two circles with centers at O and O' touch each other externally at T as shown:

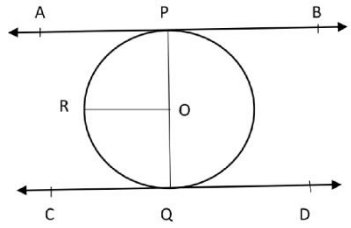


3. If PR = 12 CM, PO = 13 cm, O'Q = 5 cm and SQ = 3 cm, find the length of line segment PQ.

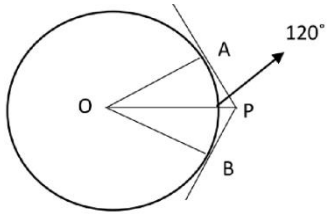
3. In the given figure, ABCD is a quadrilateral, in which ∠ADC = 90°, BC = 38 cm, CD = 28 cm and BP = 25 cm. Find the radius of the circle with center O.



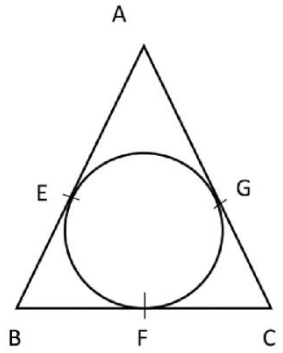
4. Prove that line segment joining the points of contact of two parallel tangents to a circle is diameter of the circle.



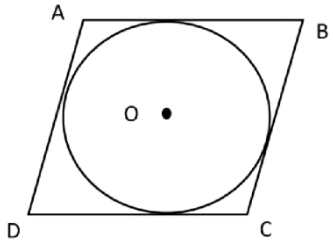
5. Two tangents PA and PB are drawn to a circle with center O, such that ∠APB = 120°. Prove that OP =AP + BP = 2AP.



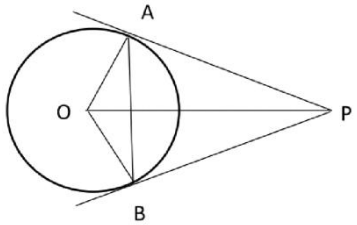
6. If the isosceles triangle ABC of the figure given below, AB = AC, show that BF = FC.



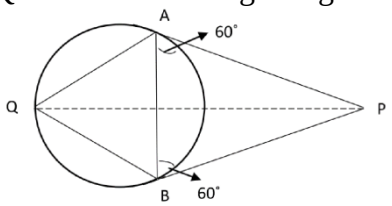
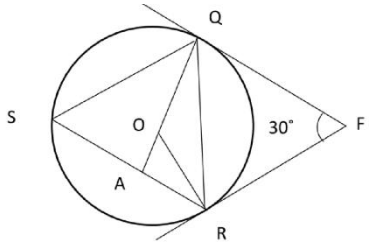
7. Prove that the parallelogram circumscribing a circle is rhombus.



8. In the given figure, OP is equal to the diameter of the circle. Prove that ABP is an equilateral triangle.



9. PA and PB are the tangents of a circle which circumscribes an equilateral ΔABQ.
If ∠PAB = 6o°, as shown in the figure, prove that QP bisects AB at right angles.



10. Tangents PQ and PR are drawn to circle such that ∠RPQ = 30°. A chord RS is drawn parallel to the tangent PQ. Find ∠RQS.